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Standard X » Mathematics » RS Agarwal

Question

Find the centre of a circle passing through the points (6, -6), (3, -7) and (3, 3).

Solution

 Verified by Toppr

Let O(x, y) is the center of the circle and A(6, -6), B(3, -7) and C(3, 3) are the points on the circumference of the circle.

$$\therefore OA = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$\Rightarrow OA = \sqrt{(x - 6)^2 + (y + 6)^2}$$

$$\Rightarrow OB = \sqrt{(x - 3)^2 + (y + 7)^2}$$

$$\Rightarrow OC = \sqrt{(x - 3)^2 + (y - 3)^2}$$

$\therefore$ Radii of the circle are equal

$$\therefore OA = OB$$

$$\Rightarrow \sqrt{(x - 6)^2 + (y + 6)^2} = \sqrt{(x - 3)^2 + (y + 7)^2}$$

$$\Rightarrow x^2 + 36 - 12x + y^2 + 36 + 12y = x^2 + 9 - 6x + y^2 + 49 + 14y$$

$$\Rightarrow -6x - 2y + 14 = 0$$

$$\Rightarrow 3x + y = 7 \dots\dots\dots(i)$$

Similarly,

$$OA = OC$$

$$\Rightarrow \sqrt{(x - 6)^2 + (y + 6)^2} = \sqrt{(x - 3)^2 + (y - 3)^2}$$

$$\Rightarrow x^2 + 36 - 12x + y^2 + 36 + 12y = x^2 + 9 - 6x + y^2 + 9 - 6y$$

$$\Rightarrow -6x + 18y + 54 = 0$$

$$\Rightarrow -3x + 9y = -27 \dots\dots\dots(ii)$$

Adding (i) and (ii)

$$\Rightarrow y = -2$$

Substitute the value of y in (i)

$$\Rightarrow 3x + -2 = 7$$

$$\Rightarrow 3x = 9$$

$$\Rightarrow x = 3$$

$\therefore$ The center of the circle is (3, -2).

Was this answer helpful?

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Q4 Centre of the circle passing through the points (6,-6), (3,-7) and (3,3) is .

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Q5 Question 3

Find the centre of a circle passing through points (6, -6), (3, -7) and (3, 3).

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